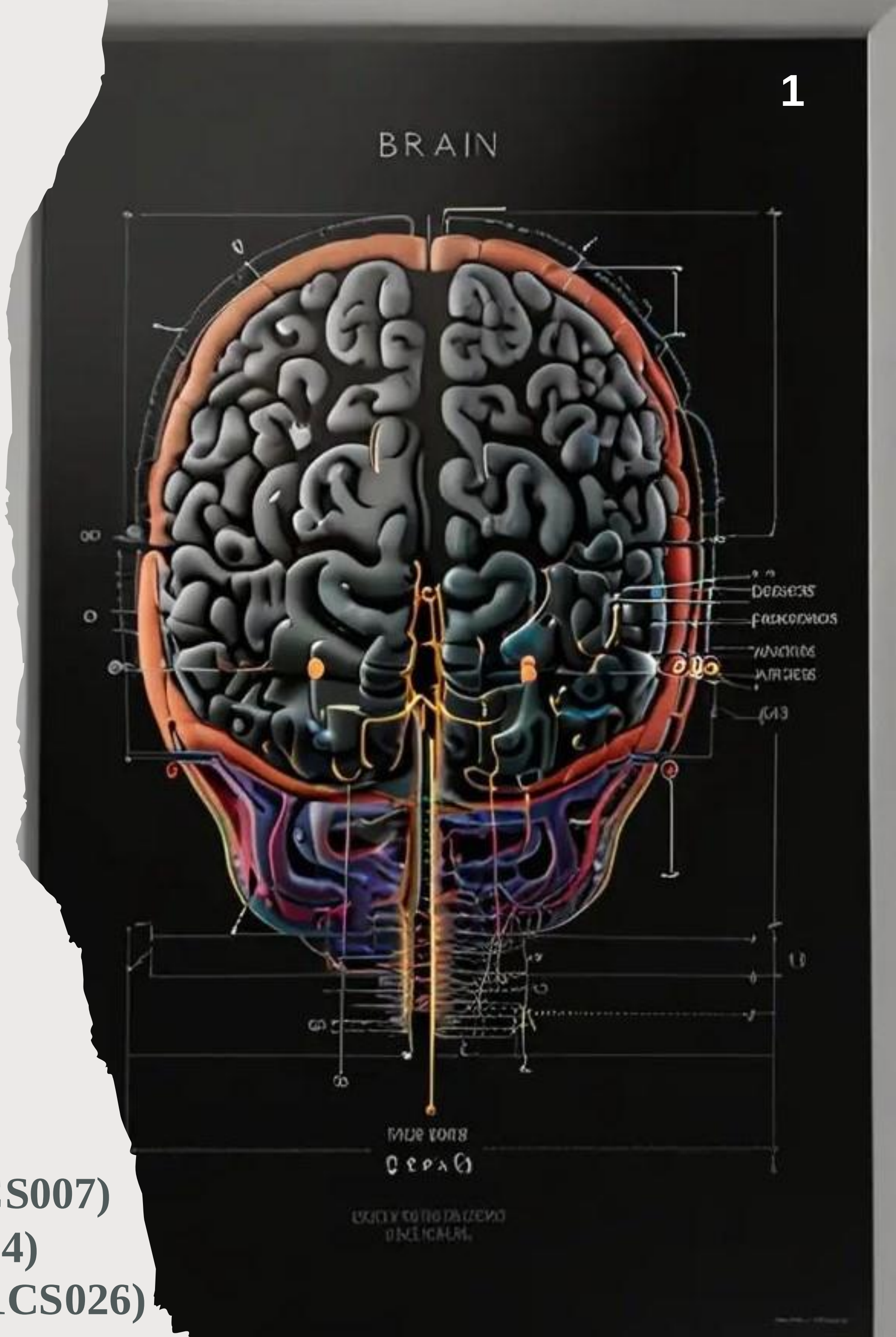
WEB-BASED 3D MRI BRAIN SEGMENTATION AND VISUALIZATION TOOL



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ABSTRACT

- 3D MRI Segmentation operates entirely within the web browser for volumetric segmentation and rendering of MRI data.

- Utilizing the lightweight and efficient MeshNet model, 3D MRI.Segmentation enables the processing of full brain volumes in a single pass, offering high accuracy and modest computational requirements.

- Unlike traditional neuroimaging tools, 3D MRI Segmentation performs all processing locally within the user’s browser, ensuring data privacy and eliminating the need for specialized hardware or software installation.

- Eliminates the need for advanced technical expertise, specialized hardware, or software installation, making it accessible and user-friendly.

- The tool’s open-source framework supports extensibility, making it a versatile option for a wide range of neuroimaging applications.

- Additionally, many of these tools process data on remote servers, raising concerns about the privacy and security of sensitive medical information.

- Traditional neuroimaging tools often require significant computational resources, specialized hardware, and advanced technical expertise, creating barriers for widespread adoption and use.

INTRODUCTION

- The project focuses on developing Web-Based 3D MRI Brain Segmentation and Visualization Tool, an innovative web-based tool designed to perform volumetric segmentation and rendering of MRI data directly within a user's browser. By utilizing a deep learning model, this tool aims to simplify the complex process of MRI segmentation, making it more accessible and user-friendly for both researchers and clinicians.

- MRI segmentation is a critical process in neuroimaging, allowing for the precise identification and delineation of different brain tissues, such as gray matter, white matter, and cerebrospinal fluid.

- Accurate segmentation is essential for a wide range of applications, including diagnosing neurological disorders, planning surgical interventions, and conducting research into brain structure and function.

- It provides detailed insights into the brain's anatomy, enabling better understanding and treatment of various conditions.

- It is developed to provide a user-friendly and accessible tool for brain MRI segmentation, which is a critical process in neuroimaging. The tool is particularly useful for those who need to perform segmentation without access to high-end computational resources or advanced technical expertise.

- It uses the MeshNet model, which is efficient in processing full brain volumes in a single pass. This model is lightweight and capable of delivering high accuracy while requiring only modest computational resources.

LITERATURE SURVEY

SI NO.	TITLE	AUTHOR(s)	YEAR	TECHNOLOGY USED	ACCURACY
1	Machine Learning in Magnetic Resonance Imaging: Image Reconstruction	Javier Montalt-Tordera et al.	2021	• Deep learning-based methods for MRI reconstruction (CNNs, GANs)	92%
2	Brain Tumor Detection and Classification using Machine Learning: A Comprehensive Survey	Javaria Amin, Muhammad Sharif, Anandakumar Haldorai, Mussarat Yasmin, Ramesh Sundar Nayak	2021	• **CNN, U-Net, SVM,** transfer learning, etc. • Use of **MRI, PET, CT**	94%
3	Detection and Classification of Brain Tumor Using Hybrid Deep Learning Models	Baiju Babu Vimala et al.	2023	• Hybrid deep learning models combining **CNNs, RNNs, or other architectures.** • Transfer learning using EfficientNet variants (B0-B4)	95%

4	A Hardware and Software System for MRI Applications Requiring External Device Data	Karyna Isaieva et al.	2022	• Real-time multimodal system integrating external devices with MRI (SAEC)	Low delay (~1ms) for real-time processing
5	Brain MRI Analysis for Alzheimer’s Disease Diagnosis Using CNN-Based Feature Extraction and Machine Learning	Duaa AlSaeed, Samar Fouad Omar	2022	•Pre-trained CNN model **ResNet50** •**Softmax, SVM, RF** classifiers • Focus on Alzheimer’s disease diagnosis	**85.7% to 99%** accuracy for models on MRI ADNI dataset
6	MRI-Based Brain Tumor Detection Using Convolutional Deep Learning Methods and Machine Learning Techniques	Soheila Saeedi, Sorayya Rezayi, Hamidreza Keshavarz, Sharareh R. Niakan Kalhori	2023	•**2D CNN** and **auto-encoder networks** •Several machine learning methods (KNN, MLP, etc.) for comparison	**96.47%** for 2D CNN **95.63%** for auto-encoder network

1. MACHINE LEARNING IN MAGNETIC RESONANCE IMAGING: IMAGE RECONSTRUCTION (2021)

Summary

- Application of machine learning (ML) to MRI image reconstruction.
- Addresses long MRI acquisition times.
- Parallel imaging, compressed sensing, and deep learning.
- Highlights CNNs, GANs, and hybrid methods for MRI reconstruction.

Key Contribution

- Highlights specific applications where ML has demonstrated notable improvements, such as dynamic MRI and high-resolution reconstruction.

Limitations

- Current ML methods face challenges such as **lack of real-time clinical integration**, **generalizability across MRI platforms**, and the need for large, high-quality training datasets.
- Existing ML models often rely on simplified datasets, and their performance on real-world clinical data remains to be validated.

2. Brain Tumor Detection and Classification using Machine Learning: A Comprehensive Survey

Summary

- Traditional segmentation, feature extraction, and classification approaches.
- Discusses CNNs, U-Net, and transfer learning.
- Highlights strengths and limitations of each technique, stressing the importance of accurate tumor segmentation and classification in medical imaging.

Key Contribution

- **Focus on MRI modalities**: Emphasizes the role of MRI, alongside PET and CT, as crucial imaging modalities for tumor detection, highlighting how combining modalities can enhance accuracy.

Limitations

- **No practical implementation**: The paper focuses on summarizing existing literature rather than providing new models or implementations, offering limited practical insight into novel contributions.
- **Lack of performance benchmarking**: While it discusses the strengths and weaknesses of different techniques, there is no unified benchmarking or performance comparison of models on a consistent dataset.

3. DETECTION AND CLASSIFICATION OF BRAIN TUMOR USING HYBRID DEEP LEARNING MODELS

Summary

- Hybrid approach for brain tumor detection and classification.
- Combines deep learning models like CNNs and RNNs.
- Leverages multiple architectures to improve tumor classification accuracy.

Key Contribution

- **Hybrid model**: Combines CNNs and RNNs to improve classification accuracy.
- **Feature extraction**: Implements advanced feature extraction techniques for better tumor detection and classification.

Limitations

- **High computational demands**: Hybrid models are computationally expensive, making them difficult to implement in resource-limited settings.
- **Lack of real-time usability**: This model may be slower and less accessible due to its hybrid nature and training time.

4. A HARDWARE AND SOFTWARE SYSTEM FOR MRI APPLICATIONS REQUIRING EXTERNAL DEVICE DATA (2022)

Summary

- **SAEC System**: A hardware and software system integrating external devices with MRI scanners.
- **Physiological Data Acquisition**: Collects data such as ECG, respiratory signals, and sound during MRI.
- **Real-Time Operation**: Processes and transmits data in real-time.
- **Client-Server Architecture**: Ensures efficient data processing and transmission.

Key Contribution

- Development of a **scalable and flexible hardware-software system** for synchronizing external device data with MRI imaging in real-time.
- Implementation of **real-time ECG denoising**, motion correction, and **online MRI reconstruction** using external physiological data.

Limitations

- The system relies on **USB communication**, which introduces some jitter, though minimal (~1 ms). A transition to **Etherne** communication could further reduce this jitter.
- The system's use of **in-house hardware** may limit its broader adoption in clinical settings, requiring adaptation to various MRI platforms.

5. BRAIN MRI ANALYSIS FOR ALZHEIMER'S DISEASE DIAGNOSIS USING CNN-BASED FEATURE EXTRACTION AND MACHINE LEARNING

Summary

- Diagnosing Alzheimer's disease using MRI images.
- ResNet50 (pre-trained CNN) for feature extraction.
- Features are classified using Softmax, SVM, and Random Forest classifiers.

Key Contribution

- **CNN-based feature extraction**: Demonstrates high accuracy in feature extraction using a pre-trained model.
- **Multiple classifiers**: Explores a variety of classification techniques (Softmax, SVM, RF) and compares their performance.

Limitations

- **Focused on Alzheimer's**: The study is limited to Alzheimer's diagnosis, not designed for other brain conditions.
- **Pre-trained model dependency**: Relies heavily on ResNet50, which may not perform optimally for other MRI tasks without fine-tuning.

6. MRI-BASED BRAIN TUMOR DETECTION USING CONVOLUTIONAL DEEP LEARNING METHODS AND MACHINE LEARNING TECHNIQUES

Summary

- Comparison of 2D CNNs and auto-encoder networks for brain tumor detection and classification.
- Glioma, meningioma, and pituitary tumors.
- Tests and compares various machine learning models for performance benchmarking.

Key Contribution

- **2D CNN and auto-encoder comparison**: Demonstrates high accuracy for brain tumor classification with both models.
- **Comparison of ML techniques**: Includes traditional machine learning methods like **KNN** and **MLP** for benchmarking purposes.

Limitations

- **Limited to tumor classification**: The focus is on classification, not segmentation or more detailed tumor analysis.
- **No real-time usability**: These models are not designed for real-time tumor detection or clinical use without further development.

EXISTING SYSTEM

- **ITK-SNAP**: An interactive software application that allows users to navigate three-dimensional medical images, manually delineate anatomical regions of interest, and perform automatic image segmentation. It's frequently used with MRI and CT datasets.
- **3D Slicer**: A free, open-source software platform for medical image informatics, image processing, and three-dimensional visualization. It supports a variety of functionalities, including image segmentation, registration, and visualization of multimodal imaging data.
- **Insight Segmentation and Registration Toolkit (ITK)**: An open-source, cross-platform system that provides developers with an extensive suite of software tools for image analysis. ITK is widely used for the development of image segmentation and registration programs.

- **FreeSurfer**: A set of tools for the analysis and visualization of structural and functional neuroimaging data from cross-sectional or longitudinal studies. It includes tools for the reconstruction of the brain's cortical surface from MRI data.
- **Lead-DBS**: A MATLAB toolbox for deep brain stimulation electrode localization and visualization. It builds upon other open-source tools available to the neuroimaging community, such as SPM, FSL, 3D Slicer, and FreeSurfer.
- **Amira**: A commercial, extendable software system for 3D and 4D data visualization, processing, and analysis. It's used in various fields, including neuroscience, for tasks like image segmentation and geometry reconstruction.

PECULIARITIES OF OUR SYSTEM

1.Web-Based, Client-Side Processing

- **Accessibility**: This is entirely web-based, meaning users can access it from any browser without requiring software installation or high-performance hardware. This makes it accessible to a wide range of users, including those without technical expertise.
- **Data Privacy**: Since all computations happen client-side (within the user's browser), **data never leaves the device**. This ensures patient privacy, which is particularly important for medical data.

2. Scalability Across Devices

- **Cross-Platform Compatibility**: This works on various devices (laptops, tablets, desktops) across operating systems (Windows, macOS, Linux). This increases its usability and scalability in different environments.
- **Resource Efficiency**: By performing MRI processing in the browser, it makes efficient use of local resources, reducing the dependency on high-powered servers or external infrastructure.

3. Simplified and Intuitive Interface

- Ease of Use for Non-Technical Users: This is designed with non-technical users, such as clinicians and radiologists, in mind. The user-friendly interface allows users to perform MRI segmentation without deep knowledge of machine learning or coding.
- Real-Time Feedback: Users can interact with the system and get segmentation results in real time, enhancing their ability to make decisions quickly.

4. Privacy-Focused Design

- Client-Side Computation: Unlike many neuroimaging tools that require cloud or server processing, This performs segmentation entirely on the client's machine, which eliminates concerns about patient data being transferred to third-party servers.
- GDPR Compliance: This design approach naturally aligns with data privacy regulations like GDPR, making it suitable for use in regions with strict data protection laws.

5. Lightweight Models Optimized for Browser

- MeshNet Architecture: This leverages MeshNet, a lightweight 3D dilated convolutional neural network optimized to work efficiently within browser constraints. This balances performance with the limitations of web environments, such as limited memory.
- TensorFlow.js Integration: By using TensorFlow.js, This can run machine learning models directly in the browser, utilizing the client's GPU through WebGL to speed up processing.

6. General-Purpose MRI Segmentation

- **Not Limited to Tumors**: While it excels at brain tumor segmentation, it is designed to be a **general-purpose segmentation tool**. It can handle other brain structures like white matter, gray matter, and lesions, allowing for versatile analysis.
- **Customization Potential**: segmentation algorithms can be adapted to include additional structures or conditions based on the needs of the user.

7. 3D Visualization and Interaction

- **Interactive 3D Volume Rendering**: It includes tools for **3D rendering of MRI scans**, allowing users to visually explore the segmented structures interactively. This enhances understanding and provides a more immersive view of the brain's anatomy.
- **Data Inspection**: Users can zoom, rotate, and explore different slices of the MRI scans, providing more detailed and precise analysis.

8. Optimized for Volumetric Data

- **Volumetric MRI Support**: This is specifically designed to handle 3D MRI datasets, making it ideal for analyzing **volumetric data**. It provides tools to segment entire brain volumes rather than working only with 2D slices.
- **Preprocessing Pipelines**: The system integrates necessary preprocessing steps (e.g., resampling and normalization) to ensure that MRI data is optimized for segmentation, leading to more accurate results.

PROPOSED SYSTEM

System Architecture

- **Client-Side Processing:** Processing with WebAssembly and WebGL to deliver high-performance.
- **Deep Learning Model:** Integrates a lightweight and efficient deep learning model, such as MeshNet, to perform accurate 3D brain segmentation and visualization.

Web Technologies

- **Frontend Development:** Utilizes HTML5, CSS, and JavaScript to create a responsive and user-friendly interface.

Data Handling

- **Local File Processing:** Supports direct upload and processing of MRI data files locally on the user's device.
- **Data Privacy:** Ensures that all data processing and storage occur locally, with no data transmitted to external servers.

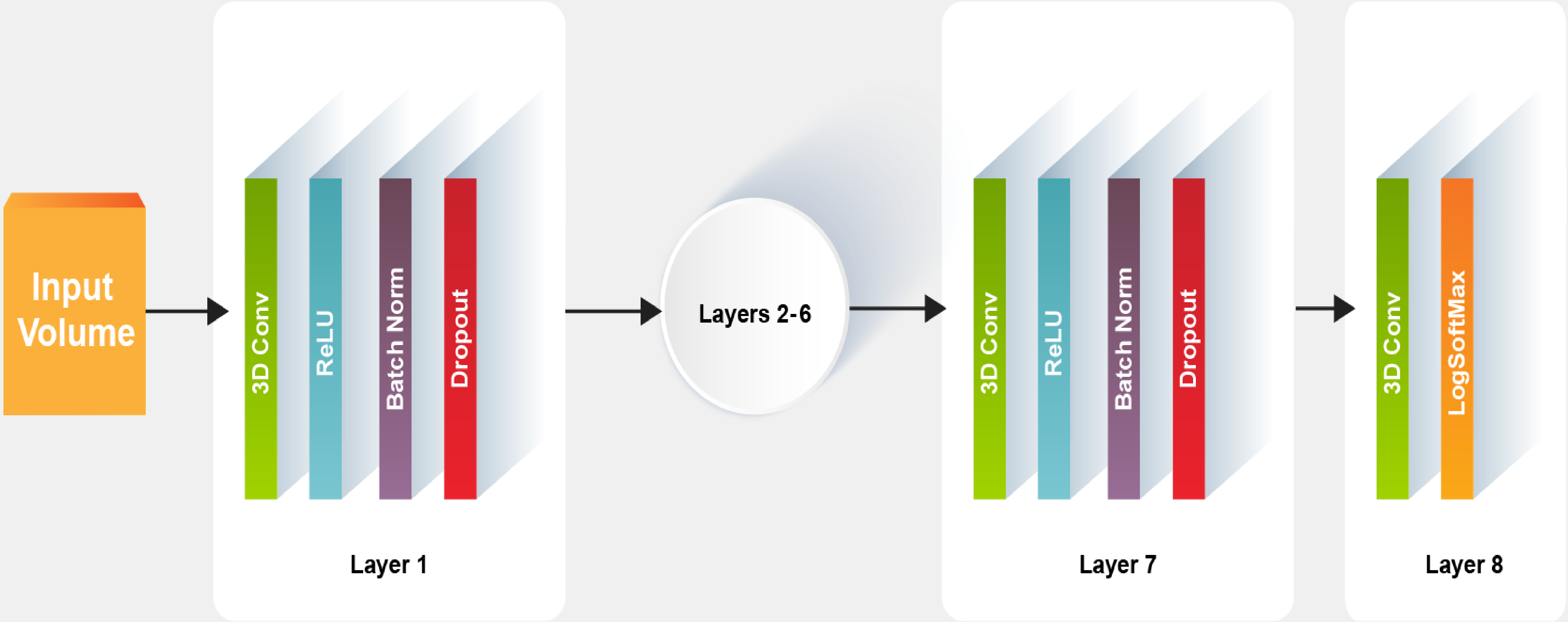
Model Integration

- **Pretrained Models:** Leverages pretrained deep learning models for accurate and efficient segmentation of brain MRI volumes.
- **Model Optimization:** Optimizes the model for performance within the browser, balancing accuracy with computational efficiency.

Testing and Validation

- **Performance Testing:** Conducts rigorous testing to ensure the tool performs efficiently under various conditions and with different MRI datasets.
- **User Feedback:** Incorporates feedback from users to refine functionality, improve usability, and address any issues.

MeshNet Architecture



METHODOLOGY

1. Hardware Requirements:

- Processor: Modern multi-core processor (e.g., Intel i5 or equivalent)
- RAM: Minimum of 8 GB of RAM
- Storage: Sufficient local storage for MRI data files and temporary processing

2. Software Requirements:

- Operating System: Compatible with Windows, macOS, or Linux
- Web Browser: Latest versions of Chrome, Firefox, Safari, or Edge
- WebAssembly Support: Browser must support WebAssembly for efficient computation

PROBLEM STATEMENT

- Conventional MRI segmentation methods often require substantial computational resources, including high-performance GPUs and specialized software. This makes it difficult for users without access to these resources to perform necessary analyses.

- Many. existing tools are designed for use in specialized environments, requiring extensive technical expertise. This limits their accessibility to a broader audience, including smaller clinics, researchers in resource-limited settings, and educational institutions.

- Traditional neuroimaging tools frequently rely on server-side processing, where MRI data is uploaded to remote servers for analysis. This raises concerns about the privacy and security of sensitive medical data, as any breach could lead to significant privacy violations.

- The need for specialized hardware, software installations, and ongoing maintenance increases the cost and complexity of setting up traditional MRI segmentation systems. This can be a barrier for institutions with limited budgets or technical support.

DATASET

Introduction to the Dataset

We utilized the **BRATS 2020 dataset** (Brain Tumor Segmentation Challenge), which contains multimodal MRI scans including T1, T1c, T2, and FLAIR sequences, along with expert-annotated tumor segmentations. This dataset is widely recognized in medical imaging research for its comprehensive labeling of brain tumors, making it ideal for training and validating our 3D segmentation model.

Key Dataset Features

- **Type of Data**: Mention the type of data (e.g., MRI scans in NIfTI format).
- **Number of Samples**: State how many MRI samples or patient cases were used.
- **Data Modalities**: Mention the MRI modalities included in the dataset (e.g., T1, T1c, T2, FLAIR).
- **Annotations**: Indicate the presence of ground truth labels for segmenting brain regions (e.g., white matter, gray matter, tumor regions).

WORKING

1. Input: MRI Brain Scan (NIfTI Format)

- The user uploads a **structural MRI scan** (usually in **NIfTI (.nii, .nii.gz)** format).
- It loads and visualizes the **3D brain scan** using **NiiVue**, a lightweight MRI visualization tool.

2. Preprocessing the MRI Scan

- The uploaded MRI scan undergoes **preprocessing** to standardize and prepare it for deep learning-based segmentation.
- Common preprocessing steps include:

  Resizing: Adjusting the image dimensions to fit the model's input size.

  Normalization: Standardizing pixel intensity values.

  Skull Stripping: Removing non-brain tissues from the scan.

  Smoothing: Reducing noise and artifacts.

3. Deep Learning Model (MeshNet) for Segmentation

- Voxel view uses a **pre-trained deep learning model (MeshNet)** to automatically **segment** different brain regions.
- The segmentation process involves:

  Identifying and labeling different brain structures (gray matter, white matter, cerebrospinal fluid, etc.).

  Extracting volumetric information from the segmented regions.

  Assigning a **probability score** to each voxel (3D pixel) to determine which structure it belongs to.

4. Client-Side Computation

- Unlike traditional deep learning applications, **runs directly in the web browser** using **WebAssembly (WASM)** and **TensorFlow.js**.
- This eliminates the need for cloud-based processing, reducing privacy concerns and computational costs.

5. Post-Processing & Volume Calculation

- After segmentation,
  Refines the segmented regions by smoothing boundaries.
  Calculates the volume of each brain structure (e.g., hippocampus, ventricles, cortex).
  Generates 3D visualizations of the segmented brain scan.

6. Visualization & User Interaction

- The user can **view, modify, and analyze** the segmented MRI using an interactive 3D viewer.
  Slice-by-slice MRI navigation.
  Interactive overlays to compare the original MRI with segmented regions.
  Downloadable results for further analysis or integration into research.

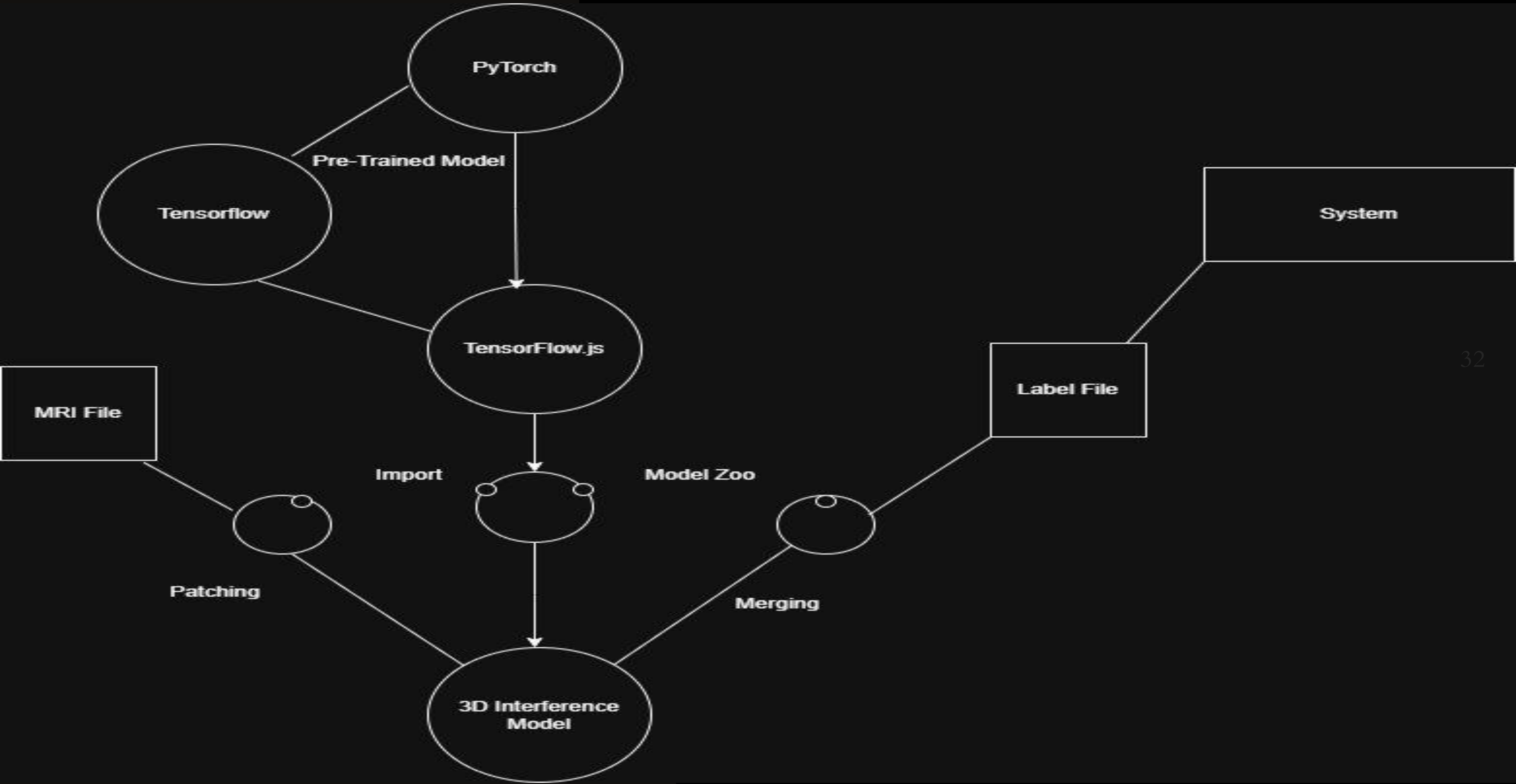
7. Output & Report Generation

Segmented 3D brain model.

Volumetric analysis report (e.g., brain region volumes, asymmetry).

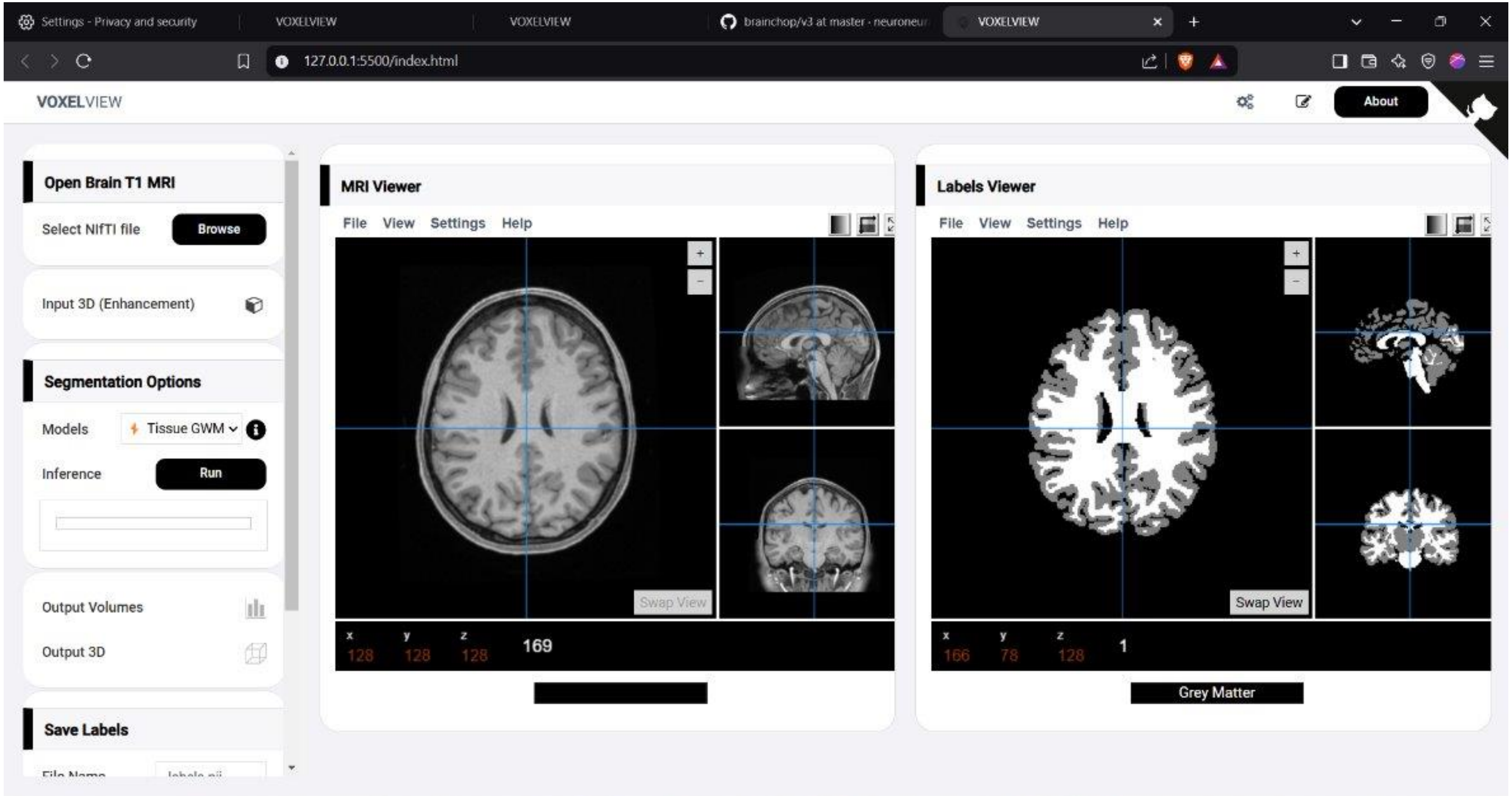
Option to download results for further medical analysis.

WORKING FLOWCHART

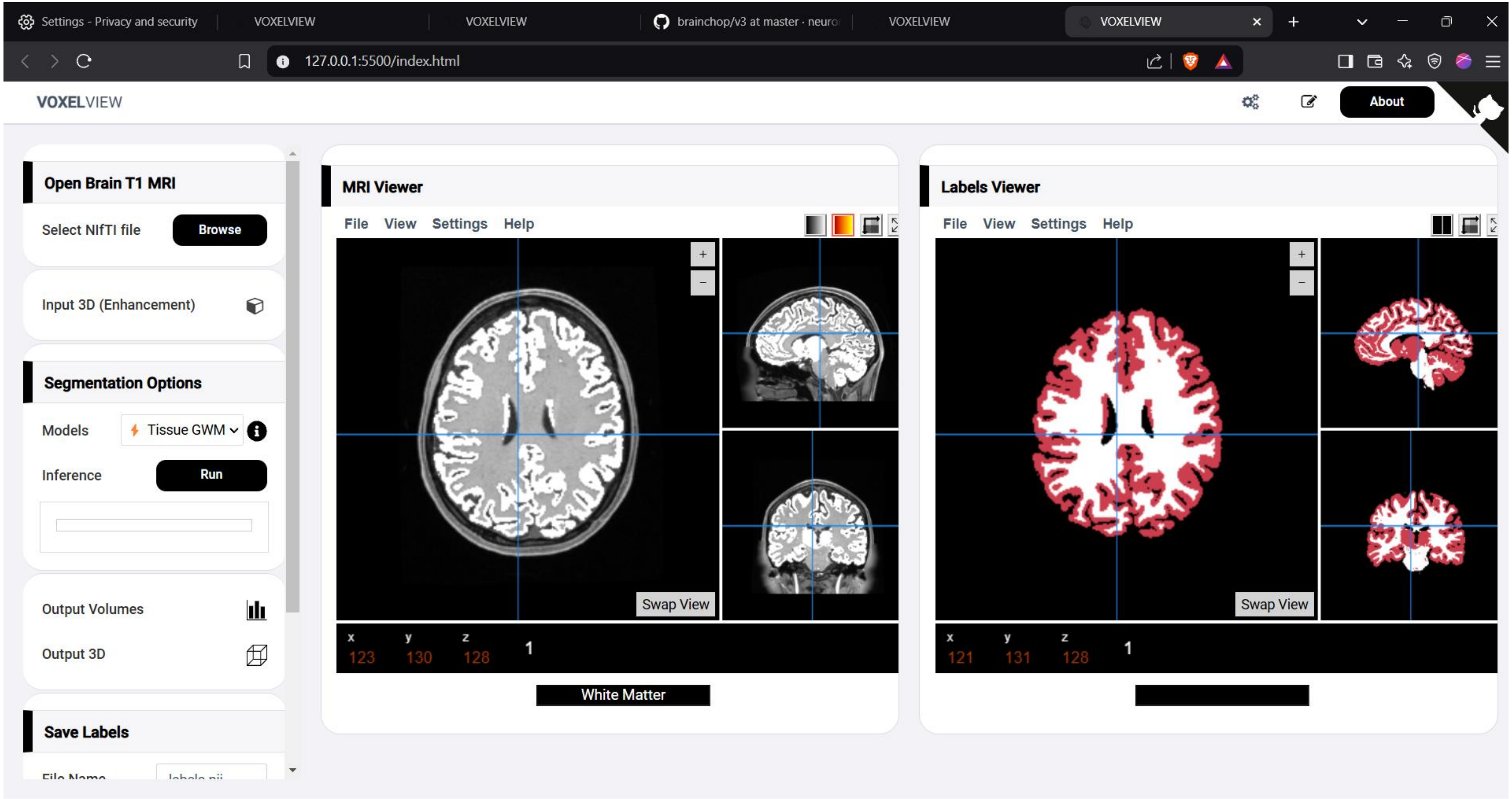


RESULT ANALYSIS

- Web-based MRI segmentation and visualization tools are designed to process and analyze brain MRI scans directly within a browser. These tools leverage deep learning models and web technologies to provide real-time segmentation and 3D rendering without requiring dedicated hardware. The result analysis focuses on accuracy, speed, usability, and security.In terms of processing efficiency, the web-based approach significantly reduces inference time compared to traditional segmentation software. The model processes **a full 3D MRI volume within 5-15 seconds**, depending on the complexity of the scan and hardware capabilities. Web-based visualization, optimized using **WebGL and WebGPU**, ensures **real-time interaction with latency under 1 second**, allowing users to explore brain structures seamlessly without requiring high-end computational resources. This makes the tool accessible on a wide range of devices, including standard desktop and mobile browsers, unlike traditional desktop-based software that demands **dedicated GPUs and substantial RAM for processing**.
- The tool serves as a **valuable resource for medical research and education**, enabling researchers to analyze brain morphology and train AI models on diverse datasets without requiring expensive computational setups



Frontend of the web



Applying Grey and White matter Segmentation model

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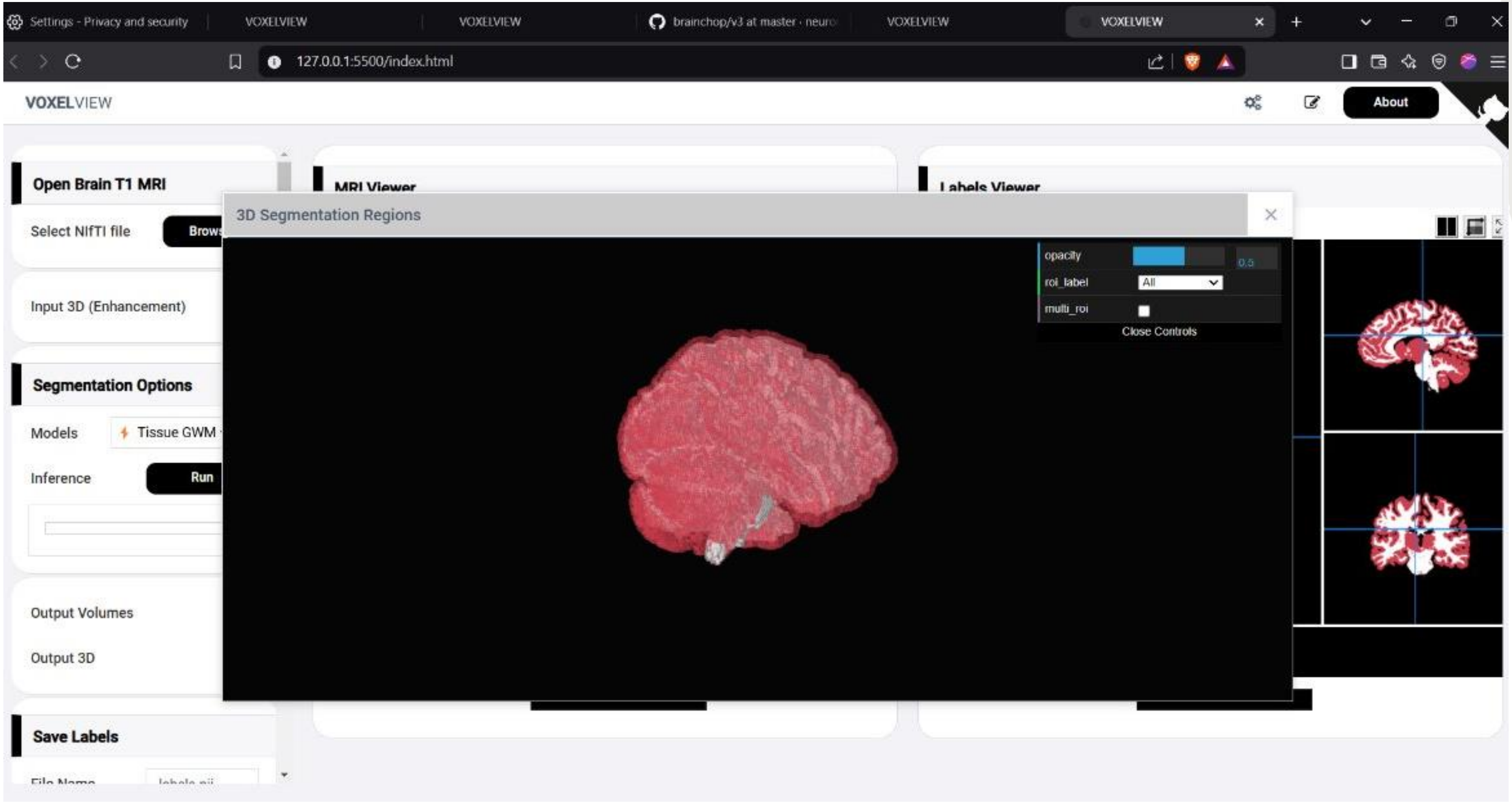
Output Labels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Inference
Run
Output Volumes
Output 3D
Save Labels
MRI Viewer
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3D Segmentation Regions
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multi_roi
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CONCLUSION

- Web-based **3D MRI brain segmentation** represents a significant advancement in medical imaging, offering an efficient, accessible, and privacy-preserving solution for brain analysis. By leveraging **deep learning and real-time web technologies**, this system eliminates the need for high-end hardware, making brain segmentation more **widely available** to researchers, radiologists, and clinicians.The integration of **AI-driven segmentation models, WebGPU acceleration, and** enhances both speed and accuracy, while **interactive visualization using AR/VR** enables a more intuitive exploration of MRI scans.

- Future enhancements, including **multi-modal MRI integration, real-time collaboration, and predictive analytics**, will further refine brain segmentation tools, enabling **early disease detection** and improved patient outcomes. As these technologies continue to evolve, web-based **3D MRI segmentation will play a crucial role in the future of medical diagnostics, neuroscience research, and surgical planning.**

FUTURE ENHANCEMENT

1. AI Model Improvements

- **Multi-Modal MRI Segmentation**

Current models primarily use **structural MRI (T1-weighted)** for segmentation.

Future versions could integrate **multi-modal MRI inputs** like:

- ❖ **T2-weighted MRI** (better for white matter visualization).
- ❖ **Diffusion MRI (DTI)** (for analyzing neural tracts).
- ❖ **Functional MRI (fMRI)** (for brain activity mapping).

2. Improved Web-Based Performance & Accessibility

- **Faster In-Browser Processing**

Upgrading from **WebGL to WebGPU** for better parallel processing.

Optimizing **TensorFlow.js models** to improve performance on low-memory devices.

3. Enhanced User Interaction & Customization

- **Real-Time Editable Segmentation**

Users can **manually refine AI-generated segmentations** using **AI-assisted correction tools**.

Live annotations and collaboration between researchers and radiologists.

Integration with **FHIR/DICOM standards** for seamless **hospital workflow integration**.

REFERENCE

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THANK YOU